

multi-Chemokine Receptor Analysis

Kilicali Isildayancan

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A0A6B0S3N8

First Domain

Sequence [66-360]: VGQIEAQLLPPLYSLVFIFGFVGNLLVVLILINCKKLKSMTDIYLLNLAIS-DLLFLLTMPFWAHYAADQWVFGNVMCKFFTGLYHIGYFGGIFFILLTIDRYLAIVHAVFALKARTVTFGVVTSGVTWVV

Homologies

Psi-blast search against Swiss-Prot:

First 5 Full Homologues

##	Receptor Name	Organism	Percent Identity	Query Cover
## 1	C-C chemokine receptor type 2	Macaca mulatta	84.07	100%
## 2	C-C chemokine receptor type 2	Rattus norvegicus	82.03	100%
## 3	C-C chemokine receptor type 2	Mus musculus	81.69	100%
## 4	C-C chemokine receptor type 5	Bos taurus	83.56	99%
## 5	C-C chemokine receptor type 2	Homo sapiens	84.48	93%
##	E value			
## 1	0.0			
## 2	0.0			
## 3	2e-179			
## 4	2e-179			
## 5	1e-174			

First 50 Unique Homologues

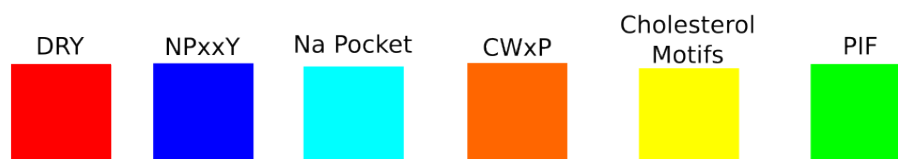
##	Receptor Name	Percent Identity
## 1	C-C chemokine receptor type 2	84.07
## 2	C-C chemokine receptor type 2	82.03
## 3	C-C chemokine receptor type 2	81.69
## 4	C-C chemokine receptor type 5	83.56
## 5	C-C chemokine receptor type 2	84.48
## 6	C-C chemokine receptor type 5	80.20
## 7	C-C chemokine receptor type 5	80.20
## 8	C-C chemokine receptor type 5	79.19
## 9	C-C chemokine receptor type 5	77.59
## 10	C-C chemokine receptor type 5	79.93
## 11	C-C chemokine receptor type 5	81.21
## 12	C-C chemokine receptor type 5	80.87
## 13	C-C chemokine receptor type 5	80.54
## 14	C-C chemokine receptor type 5	80.87

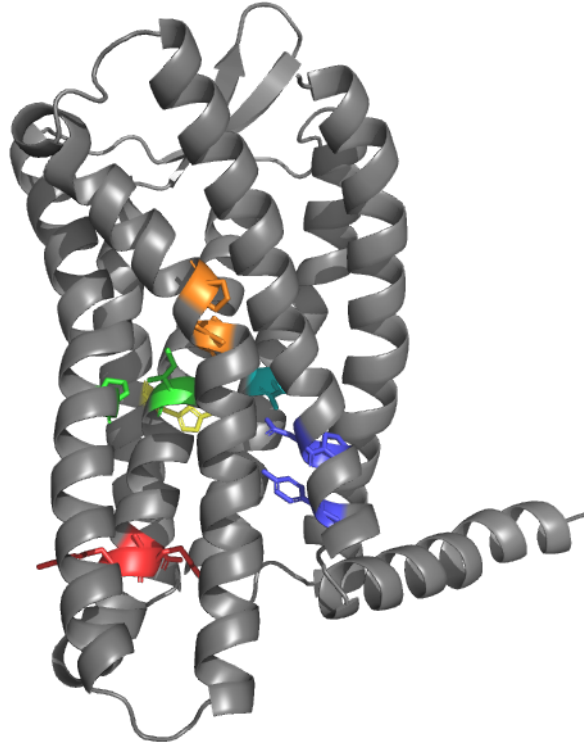
## 15	C-C	chemokine	receptor	type 5	80.54
## 16	C-C	chemokine	receptor	type 5	80.87
## 17	C-C	chemokine	receptor	type 5	80.87
## 18	C-C	chemokine	receptor	type 5	80.87
## 19	C-C	chemokine	receptor	type 5	80.20
## 20	C-C	chemokine	receptor	type 5	80.87
## 21	C-C	chemokine	receptor	type 5	80.87
## 22	C-C	chemokine	receptor	type 5	79.53
## 23	C-C	chemokine	receptor	type 5	80.87
## 24	C-C	chemokine	receptor	type 5	80.54
## 25	C-C	chemokine	receptor	type 5	77.59
## 26	C-C	chemokine	receptor	type 5	80.87
## 27	C-C	chemokine	receptor	type 5	80.54
## 28	C-C	chemokine	receptor	type 5	80.54
## 29	C-C	chemokine	receptor	type 5	80.20
## 30	C-C	chemokine	receptor	type 5	80.54
## 31	C-C	chemokine	receptor	type 5	80.54
## 32	C-C	chemokine	receptor	type 5	80.54
## 33	C-C	chemokine	receptor	type 5	80.54
## 34	C-C	chemokine	receptor	type 5	80.20
## 35	C-C	chemokine	receptor	type 5	80.20
## 36	C-C	chemokine	receptor	type 5	80.54
## 37	C-C	chemokine	receptor	type 5	79.87
## 38	C-C	chemokine	receptor	type 5	80.20
## 39	C-C	chemokine	receptor	type 5	80.54
## 40	C-C	chemokine	receptor	type 5	80.54
## 41	C-C	chemokine	receptor	type 5	80.20
## 42	C-C	chemokine	receptor	type 5	80.20
## 43	C-C	chemokine	receptor	type 5	78.86
## 44	C-C	chemokine	receptor	type 5	78.52
## 45	C-C	chemokine	receptor	type 5	78.52
## 46	C-C	chemokine	receptor	type 5	80.20
## 47	C-C	chemokine	receptor	type 5	79.53
## 48	C-C	chemokine	receptor	type 5	78.86
## 49	C-C	chemokine	receptor	type 3	61.20
## 50	C-C	chemokine	receptor	type 1	61.22

The similarity of the 50th match: 61% identity.

Structure

Motifs





Conservation

Conservation of the protein from SwissProt with highest homology to the first domain: C-C chemokine receptor type 2, UniProt ID: O18793

Conservation in the homolog

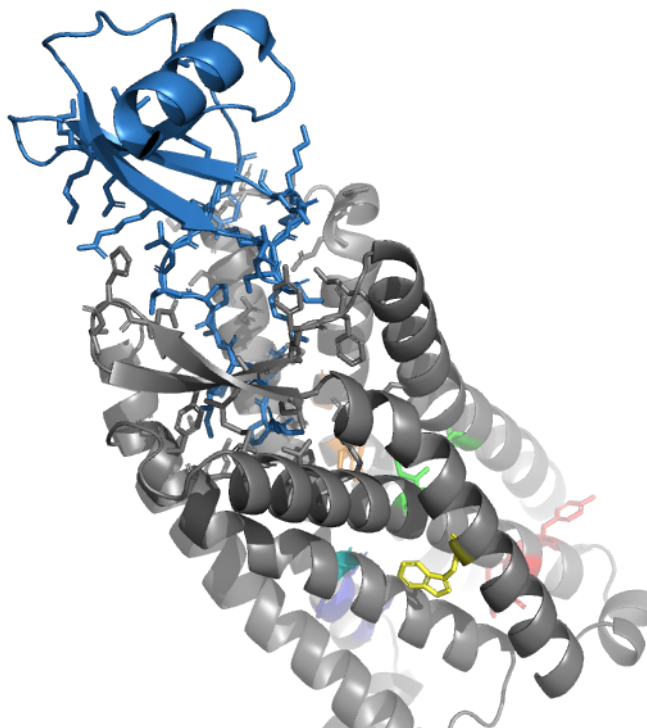
1 M L S T S R S R F I e e e e e e e e e e	11 R N T N G S G E E V e e e e e e e e e e	21 T T F F D Y D Y G A e e e e e e e e e e f	31 P C H K F D V K Q e e e b e e b e e e f
51 L V F I F G F V G N e b e e e b e e b b s s	61 M L V V L I L I N C e e b b e b b e e e f s	71 K K L K S L T D I Y b e e e e b b b b b f s s	81 L L N L A I S D L b b b b b e b b b b s s s s
101 S A A N E W V F G N e e e e e b e e e e f f	111 A M C K L F T G L Y e e b e e b b b e b s f	121 H I G Y L G G I F F b b b b b b b b b b	131 I I L L T I D R Y b b e b b e b b e e s s s s f
151 A R T V T F G V V T b e e e e e b e e b	161 S V I T W L V A V F b e e b e e e b e e f	171 A S V P G I I F T K b b e b b e e e b e s	181 C Q E E D S V Y I e e e e e e e e e e
201 F H T I M R N I L G e b b e e b b e e b	211 L V L P L L I M V I b e e b e e e b e e s s	221 C Y S G I L K T L L b b e e b e e e b e s s s s	231 R C R N E K K R H e e e e e e e e e e
251 V Y F L F W T P Y N b b b e e b e e b b s f s	261 I V I L L N T F Q E e b e e e e e e e e	271 F F G L S N C E S T e e e e e e e e e e f	281 R Q L D Q A T Q V e e b b e e b e e e f
301 N P I I Y A F V G E b b e b b e e e b e s s s s f f s f	311 K F R R Y L S M F F e b e e e e e e b e f s f	321 R K Y I T K R F C K e e e e e e e e e e	331 Q C P V F Y R E T e e e e e e e e e e
351 T A E Q E V S V G L e e e e e e e e e e f f f	361 [Empty]		

Pairwise alignment with homolog Looking at the conserved sites predicted by ConSurf, and cross-referencing to our first domain sequence:

first_domain	1	-----	
018793	1	MLSTSRSRFIRNTNGSGEEVTTFFDYDYGAPCH	
first_domain	15	LVFIFGFVGNLLVVLILINCKKLKSMTDIYLLN	
		: :	
018793	51	LVFIFGFVGNMLVVLILINCKKLKSLTDIYLLN	
first_domain	65	YAADQWVFGNVMCKFFTGLYHIGYFGGIFFIIL	
		. : . . .	
018793	101	SAANEWVFGNAMCKLFTGLYHIGYLGGIFFIIL	
first_domain	115	ARTVTFGVVTSGVTWVAVFASLPGIIFIKSLE	
		: : .	
018793	151	ARTVTFGVVTSVITWLAVFASVPGIIFTKCQE	
first_domain	165	FHTIMRSILGLVLP LLVMIVCYSGIIKTLLRCR	
		: : :: :	
018793	201	FHTIMRNILGLVLP LLIMVICYSGILKTLLRCR	
first_domain	215	VYFLFWAPYNIIVLLLSTFQEFGLSNCKSSSQL	
		. : : : :	
018793	251	VYFLFWTPYNIIVILLNTFQEFGLSNCESTRQL	
first_domain	265	NPIIYAFVGEKFRRYLSTFFRKHIAKHLCKQ--	
		: : .	
018793	301	NPIIYAFVGEKFRRYLSMFFRKYITKRFCKQCP	
first_domain	296	-----	295
018793	351	TAEQEVSVGL	360

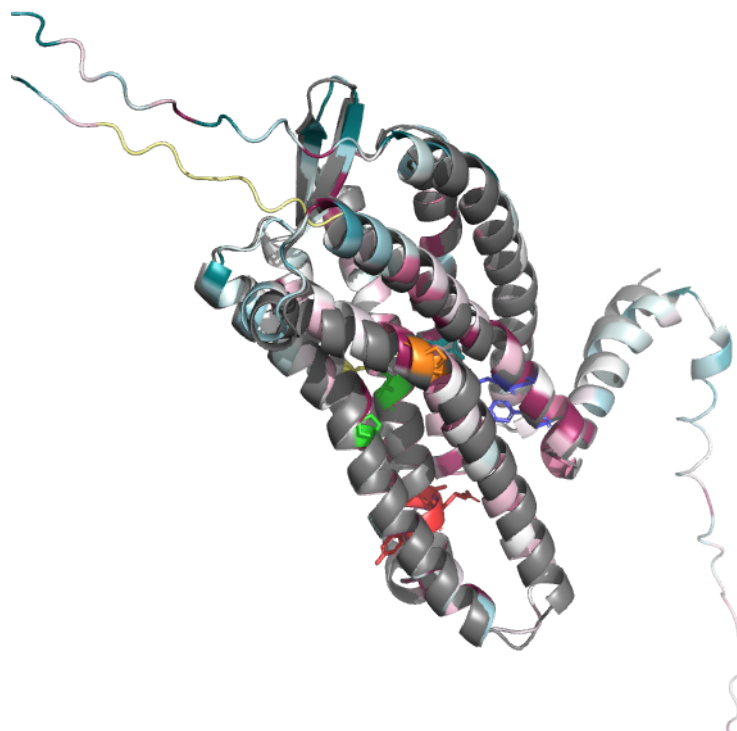
Binding ligands of homolog 7XA3 is with natural ligand (CCL2), 6GPX is a non-natural ligand (MK-0812),

Ligand superimposition of domain and homolog with ligand RMSD with the ligand-bound receptor: 1.391 Å



Of note, the extracellular side has some different amino acid residues, even though the pairwise alignment is almost identical. This might be important in deciding if the binding ligand is in fact CCL2.

Superimposition RMSD: 0.556



* Both are AlphaFold predicted structures

Second domain

Sequence [412-709]: LNVQRQAARLLPPLYSLVFIFGFVGNMLVVLILINCKKLKSMTDIYLLNLAISDLLFI-ITIPFWAHYAADQWVFGNTTCQLFTGFYFIGYFGGIFFIILLTIDRYLAIVHAVFALKARTVTFGAATSVVTWVVAVFASLI

Homologies

Psi-blast search against Swiss-Prot:

First 5 Full Homologues

##	Receptor Name	Organism	Percent Identity
## 1	C-C chemokine receptor type 5	Bos taurus	99.66
## 2	C-C chemokine receptor type 5	Allochromcebus lhoesti	87.58
## 3	C-C chemokine receptor type 5	Allochromcebus solatus	87.58
## 4	C-C chemokine receptor type 5	Cercocebus atys	87.58
## 5	C-C chemokine receptor type 5	Pongo pygmaeus	87.25
##	Query Cover E value		
## 1	100%	0.0	
## 2	100%	0.0	
## 3	100%	0.0	
## 4	100%	0.0	
## 5	100%	0.0	

First 50 Unique Homologues

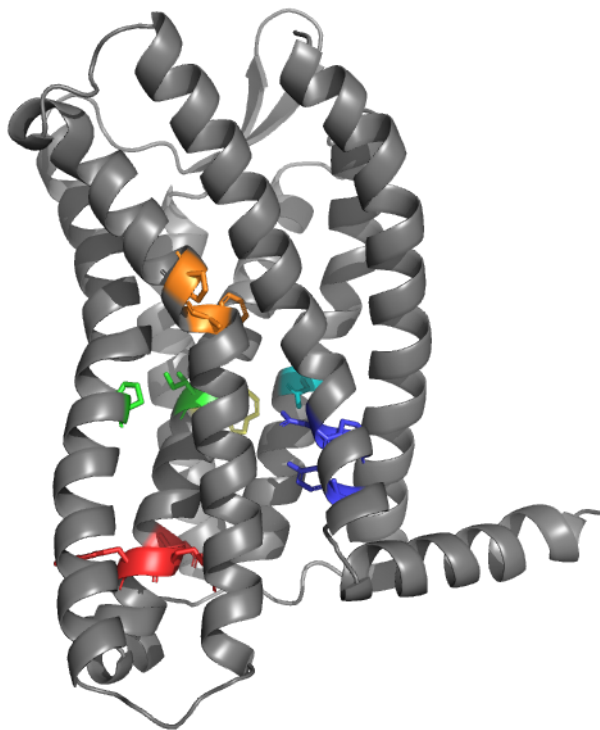
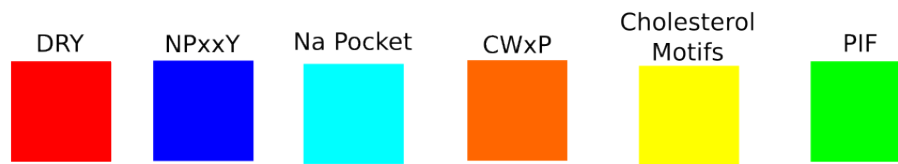
##	Receptor Name	Percent Identity
## 1	C-C chemokine receptor type 5	99.66
## 2	C-C chemokine receptor type 5	87.58

## 3	C-C chemokine receptor type 5	87.58
## 4	C-C chemokine receptor type 5	87.58
## 5	C-C chemokine receptor type 5	87.25
## 6	C-C chemokine receptor type 5	87.25
## 7	C-C chemokine receptor type 5	87.25
## 8	C-C chemokine receptor type 5	86.91
## 9	C-C chemokine receptor type 5	87.25
## 10	C-C chemokine receptor type 5	87.54
## 11	C-C chemokine receptor type 5	86.91
## 12	C-C chemokine receptor type 5	83.56
## 13	C-C chemokine receptor type 5	86.91
## 14	C-C chemokine receptor type 5	86.91
## 15	C-C chemokine receptor type 5	86.91
## 16	C-C chemokine receptor type 5	81.54
## 17	C-C chemokine receptor type 5	86.91
## 18	C-C chemokine receptor type 5	86.91
## 19	C-C chemokine receptor type 5	86.91
## 20	C-C chemokine receptor type 5	86.58
## 21	C-C chemokine receptor type 5	86.91
## 22	C-C chemokine receptor type 5	86.91
## 23	C-C chemokine receptor type 5	86.91
## 24	C-C chemokine receptor type 5	86.91
## 25	C-C chemokine receptor type 5	86.91
## 26	C-C chemokine receptor type 5	86.91
## 27	C-C chemokine receptor type 5	86.91
## 28	C-C chemokine receptor type 5	86.58
## 29	C-C chemokine receptor type 5	86.58
## 30	C-C chemokine receptor type 5	86.58
## 31	C-C chemokine receptor type 5	86.58
## 32	C-C chemokine receptor type 5	86.24
## 33	C-C chemokine receptor type 5	86.24
## 34	C-C chemokine receptor type 5	86.58
## 35	C-C chemokine receptor type 5	86.24
## 36	C-C chemokine receptor type 5	85.91
## 37	C-C chemokine receptor type 5	85.91
## 38	C-C chemokine receptor type 5	83.56
## 39	C-C chemokine receptor type 5	83.56
## 40	C-C chemokine receptor type 5	83.50
## 41	C-C chemokine receptor type 5	82.55
## 42	C-C chemokine receptor type 5	81.54
## 43	C-C chemokine receptor type 5	80.13
## 44	C-C chemokine receptor type 2	78.11
## 45	C-C chemokine receptor type 5	83.50
## 46	C-C chemokine receptor type 2	77.36
## 47	C-C chemokine receptor type 2	76.35
## 48	C-C chemokine receptor type 2	78.80
## 49	C-C chemokine receptor type 1	61.82
## 50	C-C chemokine receptor type 3	60.94

The similarity of the 50th match: 61% identity.

Structure

Motifs



Conservation

Conservation of the protein from SwissProt with highest homology to the second domain: C-C chemokine receptor type 5, UniProt ID: Q2HJ17

Conservation with homolog

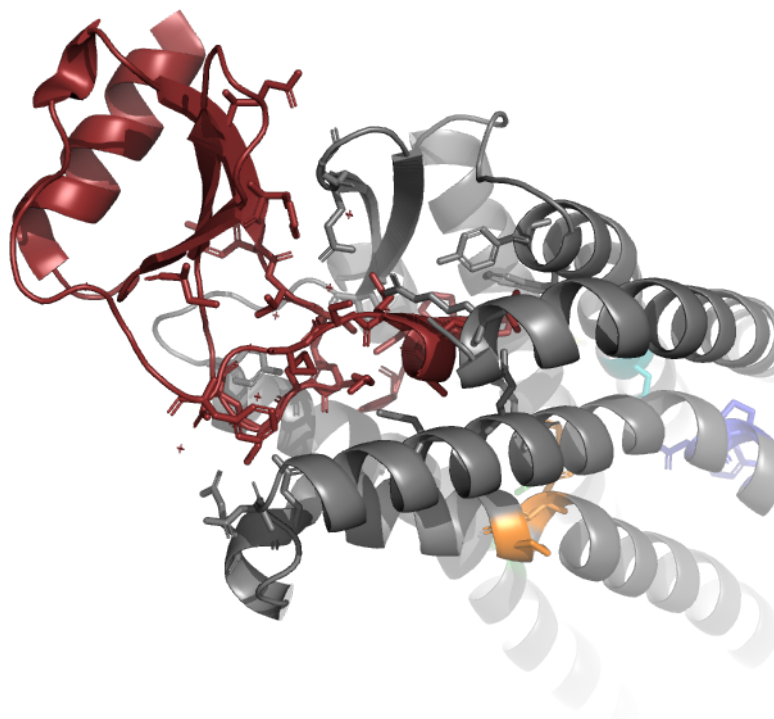
1 MDYQTSTPLY e e e e e e e e e e	11 DIDYGMSEPC e e e e e e e e e e f	21 QKLNVRQIAA e e e e e e e e b e	31 RLLPPLYSL e e b e e e b e e e
51 VVLILINCKK b b e b b e e e e e s	61 LKSMTDIYLL e e e b b b b b b b s s	71 NLAIISDLLFI b b b e b b b e b b s s s s s	81 ITIPFWAHY e b b e e b b e e e f f
101 CQLFTGFYFI b e e b b b e b b b s f	111 GYFGGIFFI b b b b b b b b b b s s	121 LLTIDRYLAI e b b e b b e e e b s s s s f f	131 VHAVFALKAA e e e e e b e e e e f f f
151 VTWVAVFAS e b e e e b e e b b f	161 LPGIIFTKSQ e e b e e b b e e e f	171 KEGSRHTCSP e e e e e b b e e e s	181 HFPSSQYHF e e e e e e e e e e
201 LGLVLP LLVM e b e e e b e e b b f s	211 IVCYSGI IKT e e b b e e b e e e s s	221 LLRCRNEKKK b e e e e e e e e e f	231 HKAVRLIFV e e e b e b b e e e f
251 YNIVLL LSTF b b e b b e e b b e s	261 QEFFGLNNCS e e e e e e e e e e f	271 GSNRLDQAMQ b e e e b e e e e e f	281 VTETLGMT H e b e b e b b b b b s f s s
301 GEKFRNYLLR b e e b e e e e e e f s f	311 FFRKYFASRF b e e e e e e e e e	321 CKGCPVFQGE e e e e e e e e e e	331 APERVSSVY e e e e e e e e e e f
351 GL e e	361 		

Pairwise alignment with homolog

second_domain	1	-----LNVRQIAARLLP
Q2HJ17	1	MDYQTSTPLYDIDYGMSEPCQKLNVRQIAARLLP
second_domain	29	VVLILINCKKLKSMTDIYLLNLAISDLLFIITIP
Q2HJ17	51	VVLILINCKKLKSMTDIYLLNLAISDLLFIITIP
second_domain	79	CQLFTGFYFIGYFGGIFFIILLTIDRYLAIVHAV
Q2HJ17	101	CQLFTGFYFIGYFGGIFFIILLTIDRYLAIVHAV
second_domain	129	VTWVVAVFASLPGIIFTKSQKEGSRHTCSPHFPS
Q2HJ17	151	VTWVVAVFASLPGIIFTKSQKEGSRHTCSPHFPS
second_domain	179	LGLVLPLLVMIVCYSGIIKTLLRCRNEKKKHKA
Q2HJ17	201	LGLVLPLLVMIVCYSGIIKTLLRCRNEKKKHKA
second_domain	229	YNIVLLLSTFQEFFGLNNCSGSNRLDQAMQVTET
Q2HJ17	251	YNIVLLLSTFQEFFGLNNCSGSNRLDQAMQVTET
second_domain	279	GEKFRNYLLRFFRKYFASRF-----
Q2HJ17	301	GEKFRNYLLRFFRKYFASRFCKGCPVFQGEAPER
second_domain	299	-- 298
Q2HJ17	351	GL 352

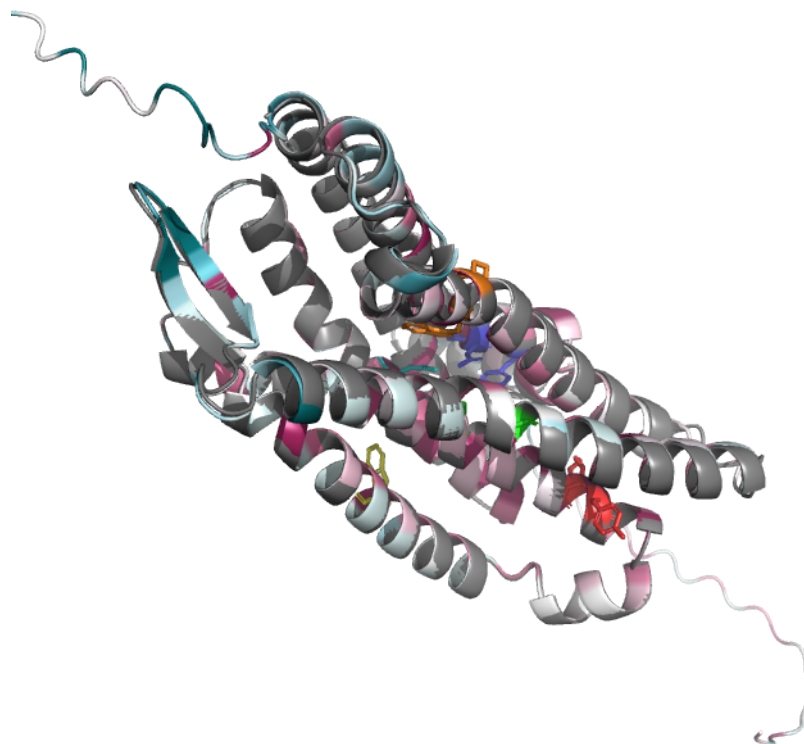
Binding ligands of homolog 5UIW is with natural ligand (CCR5), 6AKX, 6AKY is a non-natural ligand (Compound 21, 34),

Ligand superimposition of domain and homolog with ligand RMSD with the ligand-bound receptor: 0.506 Å



Of note, the extracellular side has some different amino acid residues, even though the pairwise alignment is almost identical. This might be important in deciding if the binding ligand is in fact CCL2.

Superimposition RMSD: 0.317



Pairwise alignment

first_domain	1	--VGQIEAQLLPPLYSLVFIFGFVGNLLVVLILINCKKL . . : : :
second_domain	1	LNVRQIAARLLPPLYSLVFIFGFVGNMLVVLILINCKKL
first_domain	49	AISDLLFLLTMPFWAHYAADQWVFGNVMCKFFTGLYHIC : : : .. : . .
second_domain	51	AISDLLFIITIPFWAHYAADQWVFGNTTCQLFTGIFYFIG
first_domain	99	TIDRYLAIVHAVFALKARTVTFGVVTSGVTWVVAVFASL : : : .. : .
second_domain	101	TIDRYLAIVHAVFALKARTVTFGAATSVVTWVVAVFASL
first_domain	149	HSGYACAPYFPLG---WKNFHTIMRSILGLVLP LLVMI . .: : : .. . :...
second_domain	151	GSRHTCSPHPSSQYHFWKNFQTLKIVILGLVLP LLVMI
first_domain	195	RCRNEKKKHKA VRLIFVIMIVYFLFWAPYNIVLLLSTFQ : : : :
second_domain	201	RCRNEKKKHKA VRLIFVIMIVYFLFWAPYNIVLLLSTFQ
first_domain	245	SQLDQAMQVTETLGLTHCCINPIIYAFVGEKFRRYLSTF :: : : : ..
second_domain	251	NRLDQAMQVTETLG MTHCCINPIIYAFVGEKFRNYLLRF
first_domain	295	Q 295
second domain	299	- 298

Genomic sequence

Does not exist in a genome database viewer.